CS6250 Computer Networks Syllabus
Spring 2026 Jan 12th – May 7th

Instructor Information

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General Course Information

Description

This course starts with an overview of Internet architecture and its evolution. From there, we broadly cover basic protocols and algorithms that span across all layers of the Internet protocol stack, such as TCP and congestion control. Next, we study intradomain/interdomain routing, peering and network relationships. Further, we explore router design and functionalities. Next, we study new Software Defined Networking technologies. We explore topics at the intersection of Network Security and Computer Networks, especially attacks on Internet routing such as BGP hijacking. At the final part of the course, we explore multimedia applications and Content Delivery Networks.

Hands-on projects help provide an understanding of routing, SDN, BGP hijacking, and Internet measurements. For students who would like to explore more there is an additional project that engages with tools for Internet-wide measurements to understand how large-scale events associated with unrest are reflected as connectivity disruptions for some networks in specific parts of the world.

This course begins with a brief refresher of introductory material and provides broad coverage of protocols and algorithms spanning all layers of the Internet protocol stack.

From there, we move to an overview of the Internet architecture and its evolution. We do a short refresh of basic algorithms and protocols on the transport layer, such as TCP and congestion control. Next, we study intradomain/interdomain routing, peering, and relationships amongst networks. Further, we look into router design and functionalities. Next, we study new Software Defined Networking technologies. We explore topics that intersect Network Security and Computer Networks, especially attacks on Internet routing such as BGP hijacking. At the final part of the course we look into multimedia applications and Content Delivery Networks.

The students work on hands-on projects to learn about routing, SDN, BGP hijacking, and Internet measurements. For students who would like to explore more we have designed an additional project that engages with tools for Internet-wide measurements. The students study how large scale events, that are associated with unrest, are reflected as connectivity disruptions for some networks in specific parts of the world.

Course Goals & Learning Outcomes

By the end of the course, you will:

- Learn the principles of computer networks, including signaling and framing, error control, medium access, routing, congestion control, end-to-end transport, and network APIs
- Understand, from an evolution perspective, why the Internet architecture has its current form
- Learn how basic algorithms and protocols (e.g., link-state and distance vector routing) work
- Understand problems related to convergence, instability, and configuration verification
- Understand router design and functionalities such as switching, queueing, and packet classification

- Learn about Software Defined Networking technologies and applications
- Apply Internet measurements and analytics to achieve multiple tasks
- Evaluate issues related to Internet security, surveillance, and censorship
- Understand how modern applications such as voice and video work

Pre-/Co-Requisites

A prior course in computer networking is not a prerequisite. However, this is an advanced course. Many things that would be covered in an introductory undergraduate course will not be covered in this course beyond a "review" level. Having a prior understanding of *basic* networking concepts is expected. This includes things such as IP addresses, subnets, TCP vs. UDP, OSI layer model and encapsulation, 3-way handshakes, basic operation of HTTP, client-server model vs. peer-to-peer model, etc. Students will be expected to code in Python at an intermediate level – see “programming skills” below).

Programming Skills

We will be completing the project assignments in Python 3. An intermediate level of skill with Python is adequate for the projects in this course. If you have experience learning new computer languages and feel comfortable teaching yourself a new language from scratch in a short period of time or if you have a beginner level of skill but have programmed something more complex than "Hello World" in Python before, then you should be able to learn what you need to about the language as you go through the course — it may just take you a little more time.

If you have never programmed in Python before (or have only written "Hello World" or only completed a basic tutorial on that level), then you may find it better to take some time to learn Python prior to the start of the course.

Course Materials

This class is delivered 100% on Canvas. The exams will be based on the lectures and readings found on Canvas. There is no required textbook, but Computer Networking: A Top-Down Approach, 7th edition by Kurose & Ross (ISBN 978-0133594140) is an excellent optional reference.

Classroom Management Tools are in Canvas:

- Content (Video Lectures and Reading Materials)
  - Material in Canvas -> Modules
  - Modules Summaries (presentation/recording) to be released weekly on Ed Discussion
  - Discussion in Ed Discussion
- Projects
  - Instructions and files in Canvas -> Assignments
  - Q&A in per-project TA Chat sessions and weekly office hours
  - Per-project discussion threads in Canvas -> Ed Discussion
  - Student discussions in Canvas -> Ed Discussion
- Exams
  - Delivery in Canvas/Quizzes (including proctoring via Honorlock within Canvas)
  - Per-exam logistics and discussions in Canvas -> Ed Discussion
- Grades in Canvas -> Grades

NOTE: A previous iteration of this course had lecture videos hosted on Udacity. While these can be helpful and still teach useful concepts, the old lectures on Udacity do **not** cover the same material. Students are expected to study and use the content in Canvas.

Course Virtual Machine

Projects throughout the course will be released and graded on a virtual machine (VM). SDN Firewall, BGP Hijacking, and the optional Simulating Networks projects require the use of the course Virtual Machine. Updated information about the course VM, including download links for the VM images and configuration instructions, will be available in Ed Discussion.

Course Requirements, Assignments & Grading

Assignment Distribution and Grading Scale

Grading will be based on five projects and two exams completed throughout the course, with an extra credit opportunity. The grade breakdown is as follows:

Assignments	Weight
Spanning Tree Protocol	8%
Distance Vector Routing	14%
SDN Firewall	14%
BGP Hijacking	15%
BGP Measurements	15%
Projects Total:	**66%**
Exam 1	12%
Exam 2	12%
Quizzes	10%
Exam/Quizzes Total:	**34%**
Extra Credit: Internet Wide Events **-or-** Pathway to Research	3%
Extra Credit Total:	**3%**

Note: The Internet-Wide Events extra credit submissions will only be graded if your course grade is within three (3) points of the next higher letter grade. All Pathway to Research submissions will be graded. If you complete both extra credit assignments, the highest score will be used to calculate your course grade.

Grades

There is no rounding or curving in CS 6250. Letter grades are calculated as follows:

- **A** ≥ 90%
- **B** ≥ 80% and < 90%
- **C ≥** 70% and < 80%
- **D** ≥ 65% and < 70%
- **F** < 65%

Feedback on Exams and Projects

Discussion of each exam will be allowed once all students have submitted the exam and grading has been completed. We release grades pending Honorlock review. **It is possible for you to see a score for an exam and later have that grade updated or changed to "EX" if our review shows you did not follow the proctoring rules or if the review indicates academic misconduct.**

For projects submitted to Canvas, your last submission on Canvas will be used for grading. Be sure to include ALL files in the last submission. For projects submitted on Gradescope, your last submission or the submission you mark as "Active" will be used for grading.

We will not release full solutions to any exam or project.

Course Components

- **Lectures & Readings:** The lectures and accompanying readings for each week. Students are responsible for watching and reading this material.
- **Projects:** Project start and end dates are provided on the schedule at the end of this Syllabus. Note that instructions and project files are provided in Canvas.
- **Quizzes:** There are weekly quizzes for each week lesson content is assigned. These are due at the end of the week -- there are no exceptions for people who don't take Quizzes on time.
- **Exams:** Exams will be administered via Honorlock. Please check the class schedule for open/close dates of the exam window.
    - Late exam submissions, after the window has closed, are not allowed – except in case of emergency
    - You must take all exams by yourself, and all answers must be provided entirely by you
    - Exams do not allow supplemental materials (notes, scratch paper, books, or digital reference)
    - Only the Honorlock calculator app is allowed
    - **See detailed proctoring requirements below for complete guidelines and penalty information**

Proctoring Information

All course exams will be proctored via Honorlock. A proctored exam is similar to the one you would take in the classroom. This means no open textbooks, notebooks, notes, and other like resources are allowed.

Honorlock uses multi-factor biometric authentication to verify identity. Each student will provide face and ID scans, which will be measured against the student's stored baseline biometric profile. You will also be asked to scan the room around you – see below and the pinned post on Ed Discussion for details.

You will have the opportunity to take an onboarding quiz to become familiar with how it all works and to ensure that your system meets all hardware and software requirements. The onboarding quiz will be a practice quiz that will not affect your grade in the course. You can take the onboarding quiz as many times as you like. All potential violations are reviewed by a human - **a violation can result in a grade of 0 for an exam and/or point deductions**. If you have any issues with Honorlock while taking the graded exam, reach out to Honorlock 24/7. Support: https://honorlock.com/support/. TA support may be unavailable; especially as the exam window closes.

It is critical that you perform a comprehensive, high-quality room scan. **An insufficient room scan will result in an automatic 10-point deduction.** It is also critical that you ensure there are no prohibited items in the testing area. **The presence of any prohibited items will result in an automatic 10-point deduction.**

Room scan requirements – failure to do any of the following will result in an automatic 10-point deduction:

- Show a clear view of the entire testing area
    - Perform a 360-degree scan of your surroundings to show that your **testing area is clear of notes and unauthorized materials**.
    - Move the camera/mirror slowly and smoothly.

  - **Show your monitor and keyboard are clear of notes and unauthorized materials.**
  - After completing your room scan, review the video to ensure that it's easy to see the whole area.
- Webcam Requirements:
  - If you use a **laptop webcam** to conduct the room scan, you **must use a mirror** to show what will be in your eyeline during the exam. The mirror must be at least 6" x 10" and bigger is better. Do not use your cell phone as a mirror.
  - If you use an **external webcam** to conduct the room scan, you must also use the same external webcam during the exam - do not switch to a laptop webcam after performing the room scan.

Testing area requirements – the presence of any of the following will result in an automatic 10-point deduction:

- **Extra Monitors**. Extra monitors that are not turned to face away or covered by a towel or other material are not permitted. If you use an external monitor for your laptop, you must show that the laptop is closed.
- **Cell Phones, Smart Watches**, **Tablets or Other Devices with a Screen**. No device with a screen is permitted in the exam area.
- **Headphones or Earbuds**. If you need to control environmental noise, earplugs are permitted. Any device that transmits sound is not permitted in the exam area (or in your ears).
- **Written Material**. Written material, including papers, books, notes, receipts, flyers, or even junk mail, is not permitted in the exam area. This applies even to blank paper and materials not related to CS 6250.
- **Other People**. It is not permitted to have other people around you during the exam.

Assignment Due Dates

All quizzes, exams, and projects are due at 23:59:59 AOE, unless otherwise noted. All assignments are due relative to the "Anywhere on Earth" Time Zone (AOE). We will not make exceptions due to time zone issues. If your time zone is set correctly in Canvas, all dates and ttimes will be accurately reflected in your chosen tome zone. **It is your responsibility to make sure your time zone is set correctly in Canvas!**

Technology Requirements and Skills

Minimum Technical Requirements

- **Browser and connection speed:** An up-to-date version of Chrome or Firefox is strongly recommended. 2+ Mbps recommended; at minimum 0.768 Mbps download speed.
- **Operating system:** Note – these technical requirements are for project completion only. Honorlock requires the Chrome browser with the Honorlock extension installed, on Windows/Mac only.
    - PC: Windows 10 with latest updates installed
    - Mac: OSX Sierra 10.12 or higher with latest updates installed
    - Linux: Any recent distribution that has the supported browsers installed
- **Hardware:** Georgia Tech's online graduate program issues the following Minimum Hardware Requirements to online grad students (https://omscs.gatech.edu/technical-requirements). We recommend that you meet or exceed these guidelines to ensure you have sufficient computing power to complete all coursework and projects. You need **8Gb RAM** minimum to run a VM. Additionally, **M1 computing machines** will have limited support from the TA staff.
- **Software:** We will have limited support for VMware, VirtualBox, and Parallels. See Ed Discussion for other options, discussions and VM support.

Technical Support

For any technical questions, problems or concerns with lecture videos, Ed Discussion, Canvas, or other Georgia Tech IT resources please find email contacts below. The instructor/TAs will likely not be able to assist you with this.

- For OIT (Georgia Tech IT dept.) support, please email support@oit.gatech.edu
- For technical support with Canvas please email: support@instructure.com
- For technical support with Ed Discussion, please email: support@edstem.org

Canvas & Kaltura

This class will use Canvas and Kaltura to deliver course materials to online students. ALL course materials, assignments, and exams will take place on Canvas.

Course Policies, Expectations & Guidelines

Communication Policy

All communications for the class will take place within Ed Discussion. This includes, but is not limited to the following:

- **Student Questions:** Students are strongly encouraged to post their questions on Ed Discussion related to the lectures, readings, weekly discussions, exam preparation, or projects. Due to the large volume of this class, we do not recommend directly emailing the instructor or the TAs.
- **Private Posts:** Students can post privately to the teaching staff on Ed Discussion. This is appropriate when a student needs to ask a question about a personal matter or request a regrade. Students may also post privately to ask questions about material when the question would violate academic integrity rules if posted publicly. An example of this would be asking a question about their code submission for a project that requires posting the code. Posts should be addressed to "Instructors" to ensure that they are seen and addressed.
- **Volume of posts:** Please try to avoid posting duplicate questions. Posting duplicate questions can lower the turnaround time of responses significantly. You can use the search bar (to the right of the "New Post" button) and the folders listed across the top of the page to look for related posts where your question may have already been asked & answered.
- **Instructor/TA Announcements:** Announcements within Ed Discussion will be used to communicate updates to projects, grades being posted, and other administrative information.

Students should be active on the Ed Discussion forums, and are responsible for reading announcements within 24 hours, as the information typically will be time sensitive. While the teaching staff will make every effort to update resources/descriptions on Canvas in the event of a policy or project change, it is ultimately the responsibility of the student to obtain updates on Ed Discussion. This includes all posts, whether they have been pinned by the teaching staff or not.

- **Email announcements through Ed Discussion:** Particularly important announcements may occasionally be sent by email. We will use Ed Discussion to do this, so you will receive these announcement emails at the email address you have in your Ed Discussion account. We use your Canvas / GA Tech email address to add you to Ed Discussion – you can then configure another email address in Ed Discussion. However, whichever you use, you are responsible for checking it daily in case of such announcements.
- **Ed Discussion code of conduct:** Please review the anti-harassment policy, located within this document.
- **Emailing the Instructor:** Students may directly email the instructor if there is an issue that has not been resolved through communication on Ed Discussion. Due to the large volume of students, direct emails may take longer to be answered. Students are strongly encouraged to post on Ed Discussion first and seek resolution from the TA team. Emails to the Professor will likely be referred to the head TAs.
- **Other Official emails:** Under certain circumstances, the TA team may send you direct emails to your official Georgia Tech email address (you may not request a different email address for these messages). Examples include resolving academic integrity violation incidents, notification of incomplete onboarding with proctoring software, etc. Students are expected to check their Georgia Tech email at least twice a week and to respond in a timely manner.

Anti-harassment policy:

Please follow these guidelines to ensure that everyone has a beneficial, positive and harassment-free experience:

- Any type of hostile behavior will not be accepted. Abusive/degrading/hostile/intimidating language, language that creates discomfort, or interferes with a person's participation or opportunity for participation will not be tolerated. Excuses such as “It was obvious” or “I was just joking” will not be accepted.
- Students that are violating this policy may be excluded from participation at Ed Discussion. If you believe that you or a classmate has been harassed, please contact the Instructor/TAs immediately.
- Please remember that on the other end of a posted question, follow-up, or comment, there is a real person - like you - who will react not only to the content of your response but also to the tone. Please remember that students in this class have a wide variety of backgrounds and prior experience. Students with more experience or background in a topic are strongly encouraged to share - through useful and practical responses – in a supportive way so that others can learn as well. When you answer a question, please remember that your primary purpose is to help the student who asked.
- You are encouraged to suggest ways to improve questions that are posted, so that students will receive faster and better answers. You are expected to do so using positive and supportive language and tone.

Subject to Change Statement

The syllabus and course schedule may be subject to change. Changes will be communicated via email, Ed Discussion, and/or the Canvas announcement tool. It is the responsibility of students to check email messages and course announcements to stay current in their online courses.

Office Hours

The tentative office hours schedule for **Spring 2026** is as follows:

- **Weekly Office Hours with Head TAs:** These sessions are mostly focused on project questions.
- **Chat Sessions:** In addition to the above, the class TAs will hold 30 min chat sessions, 1 to 5 times per week, to help students with project questions. We will be announcing the schedule of the chat sessions every week, depending on TA availability. There are no chat sessions during Exam weeks.
- **Content Threads:** We have weekly content threads for the students to ask content questions.

Student questions in office hours: We expect students to ask questions on the Ed Discussion Office Hours thread each week. If the students are unable to watch at the time of the office hours, they are strongly encouraged to post their questions in advance and watch the recording of office hours.

Format: We will be holding office hours online using Microsoft Teams. Meetings will be listed on Canvas under the Microsoft Teams link and students are encouraged to post their questions as follow-ups to the office hours Ed Discussion post both before office hours begin and during the live broadcast. After office hours have been completed, a recording of the office hours will be made available. By entering questions in advance and viewing the recording afterward, students may participate in office hours even if they are not able to attend live.

University Use of Electronic Email

A university-assigned student e-mail account is the official university means of communication with all students at Georgia Institute of Technology. Students are responsible for all information sent to them via their university-assigned e-mail account. If a student chooses to forward information in their university e-mail account, he or she is responsible for all information, including attachments, sent to any other e-mail account. To stay current with university information, students are expected to check their official university e-mail account and other electronic communications on a frequent and consistent basis. Recognizing that some communications may be time-critical, the university recommends that electronic communications be checked minimally twice a week.

Plagiarism & Academic Integrity

Students are expected to follow the Georgia Tech Honor Code (https://osi.gatech.edu/students/honor-code), including the Graduate Addendum. All incidents of suspected dishonesty will be reported to and handled by the Office of Student Integrity.

AI/LLM Prohibition: The use of any artificial intelligence, large language models, or AI-assisted coding tools is strictly prohibited in this course. This includes but is not limited to

- ChatGPT, Claude, Gemini, or any conversational AI
- Github Copilot, Claude Code, Cursor, or any AI code completion tools
- Any other AI-powered assistance for coding, writing, or problem solving

Any violations of this policy will be considered academic misconduct and subject to disciplinary action

Collaboration: With the exception of the Pathway to Research extra credit project, where groups are allowed and encouraged, you are to complete all projects yourself. You may discuss the projects with your classmates, but you may not share code/data/solutions with other students. For specific projects, we provide explicit instructions about what pieces of information can or cannot be shared with the

entire class. Should a classmate send you their code (such as in a direct message on Slack), it is your responsibility to warn them and/or inform the TA team.

Plagiarism: You may not copy or reference solutions (fully or in part) from any source. This includes solutions from previous classes or solutions in other programming languages. Each file you turn in must be written entirely by yourself. If it is a file provided with the project, the modifications to the file must be made entirely by yourself. For written reports, all quotes must be clearly cited.

Publishing exam/project material or solutions: With the exception of private messages to the TA team, you may not publish or provide project solutions on any medium at any time, even if the solution is not working/did not earn full credit. You may not publish or provide exam questions OR answers on any medium at any time (except for in the clearly designated thread on Ed Discussion after the TA team announces grades). These restrictions include public git repos, sites such as CourseHero, and private messages to others. ***These restrictions apply even after completing the course and/or graduating from the program***.

Accommodation for Students with Disabilities

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services at (404)894-2563 or http://disabilityservices.gatech.edu/, as soon as possible, to make an appointment to discuss your special needs and to obtain an accommodation letter. Please also send a private message to "Instructors" on Ed Discussion as soon as possible. Note: the TA team cannot provide any accommodation or extensions without an accommodation letter, nor are accommodations provided retroactively.

Late and Make-up Work Policy

All due dates are 23:59:59 AOE, on the date specified, unless otherwise noted. Misunderstanding of the deadline and late policy is NOT a valid justification for an extension or reduction in penalties. It is the student's responsibility to accommodate for technical issues, such as a slow connection to Canvas.

- **Late project submissions will be penalized 5% of the max point total per hour late, with the hour rounded up.** Submissions will be accepted up to 19 hours after the posted due date. If there are multiple submissions, the last submission is used for grading, ***with any late penalty applied.***
- Exams are only available and accepted during the posted exam window.
- Extra credit submissions are **NOT** accepted past the posted deadlines.

Extensions to Deadlines

Extensions to deadlines will be allowed only in the case of a medical or personal emergency, or in the case of military exercises/deployments for service members. These situations must be approved by the office of Dean of Students. Please contact the Office of the Dean of Students, (see the Division of Student Life [studentlife.gatech.edu] and the "Get Help Now" link) and send a private message to the TA team on Ed Discussion as soon as possible, letting us know of your situation.

Regrade Requests

- **Format:** All requests for a regrade on any project or other questions regarding your grade must be made via private post to all instructors on Ed Discussion. Please use the provided regrade request tag in Ed Discussion for project regrade requests. We post instructions in each grade release post. For questions about the assignment not specific to your submission, please make a public post on Ed Discussion.
- **Deadline for regrade requests:** Regrade requests must be made no later than one week from the grade release date for that assignment. Due to the volume of this class, there will be no exceptions.

- **Valid requests:** Please submit a request if you think there has been an error in grading or applying the rubric (such as the number of points not adding up correctly). Other regrade requests will not be considered: e.g. you need a higher grade for Institute/program requirements or reimbursement, your grade is close to the cutoff for the next grade level, etc.
- **Volume of regrade requests:** Please submit a regrade request only if there has been an error. Please refrain from submitting multiple requests, as they consume significant amounts of TA time. This can result in slower turnaround time that may affect the entire class.

Student-Faculty Expectations Agreement

At Georgia Tech, we believe that it is important to strive for an atmosphere of mutual respect, acknowledgement, and responsibility between faculty members and the student body. See http://www.catalog.gatech.edu/rules/22/ for an articulation of some basic expectations that you can have of us and that we have of you. In the end, simple respect for knowledge, hard work, and cordial interactions will help build the environment we seek. Therefore, we encourage you to remain committed to the ideals of Georgia Tech while in this class.

CS6250 Computer Networks Syllabus
Spring 2026 Jan 12th – May 7th

Course Schedule

Week/Dates	Topic	Assignments
Week 1 **Jan 12 – Jan 18**	Registration Week Lesson 1: Introduction, Internet History, and Architecture **Registration through Jan 16, 11:59 AM EDT**	Assigned: Simulating Networks (Optional) Assigned: Spanning Tree Protocol Assigned: Pathway to Research EC
Week 2 **Jan 19 – Jan 25**	Lesson 2: Transport and Application Layers	Due: Lesson 1 Quiz Due: Lesson 2 Quiz
Week 3 **Jan 26 – Feb 1**	Lesson 3: Intradomain Routing	Due: Lesson 3 Quiz Due: Spanning Tree Protocol
Week 4 **Feb 2 – Feb 8**	Lesson 4: AS Relationships and Interdomain Routing	Due: Lesson 4 Quiz Assigned: Distance Vector
Week 5 **Feb 9 – Feb 15**	Lesson 5: Router Design and Algorithms: Part 1	Due: Lesson 5 Quiz
Week 6 **Feb 16 – Feb 22**	Lesson 6: Router Design and Algorithms: Part 2	Due: Lesson 6 Quiz Due: Distance Vector
Week 7 **Feb 23 – Mar 1**	Exam Preparation	
Week 8 **Mar 2 – Mar 8**	Exam 1	Exam 1
Week 9 **Mar 9 – Mar 15**	Lesson 7: SDN Part 1	Due: Lesson 7 Quiz Assigned: SDN Firewall
Week 10 **Mar 16 – Mar 22**	Lesson 8: SDN Part 2	Due: Lesson 8 Quiz Due: SDN Firewall
Week 11 **Mar 23 – Mar 29**	Lesson 9: Internet Security	Due: Lesson 9 Quiz Assigned: BGP Hijacking
Week 12 **Mar 30 – Apr 5**	Lesson 10: Internet Surveillance and Censorship	Due: BGP Hijacking Due: Lesson 10 Quiz
Week 13 **Apr 6 – Apr 12**	Lesson 11: Applications Part 1: Video	Due: Lesson 11 Quiz Assigned: BGP Measurements Assigned: Internet-Wide Events EC
Week 14 **Apr 13 – Apr 19**	Lesson 12: Applications Part 2: CDNs	Due: Lesson 12 Quiz
Week 15 **Apr 20 – Apr 26**	Lesson 13: Future of the Internet (Optional)	Due: BGP Measurements Due: Internet-Wide Events EC Due: Pathway to Research EC
Week 16 **Apr 27 – May 3**	Exam 2	Exam 2 May 12: Grades Available